Unraveling the Manifold of Intelligences

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1 Motivation

This agenda aims to stay at the frontier of what may be the next large jump in AI capabilities. Several industry leaders now say that scaling current models will not be enough on its own and that one or two further breakthroughs on the scale of the transformer are still needed to reach AGI: Sam Altman (The Decoder 2023; Marcus 2026), Demis Hassabis (Axios 2025), Ilya Sutskever (Dwarkesh Podcast 2025) and Yann LeCun (The Decoder 2025). Foundational architectures of AI were largely shaped by neuroscience, e.g. the perceptron was proposed as a model of the brain (Rosenblatt 1958), Hopfield's recurrent networks (Hopfield 1982), convolutional networks (Fukushima 1980), etc. Though the biological substrate proved to be a fruitful means for the development of models of intelligence we believe that there's a much more fundamental space to be explored in order to understand the underlying characteristics that make systems intelligent. Transformers are a great example of how moving away from biological systems yielded a better result for generalizing intelligence (Vaswani et al. 2017).

2 Charting the Space of Intelligent Systems

Rather than choosing between computational and biologically inspired approaches, we propose to chart the space of possible intelligent systems. Disagreeing with Penrose's theory that intelligence requires a mind or consciousness (Penrose 1994, 2022), we believe that intelligence requires neither, though we do agree that in order to understand its principles one needs to be open to much broader explorations than the ones currently being considered. In order to conduct such explorations we intend to use tools from information theory and causal inference, as well as a myriad of tools from intersecting areas that are exploring what intelligence is in their own domain. In this context it is also important to define intelligence; here we use Levin's definition: *intelligence is competency in navigating a problem space, reaching the same goal by different means* (Levin 2022; Fields & Levin 2022, after James 1890). Here, intelligence is a multi-dimensional spectrum that might range from a sorting algorithm adapting as its environment changes, to a regulatory system in a biological body, to the resolution of hundred-year-old problems through in silico computation.

3 Information-Theoretic Tools

To map this landscape we intend to use information-theoretic tools. These tools offer a mathematical grounding for measuring complexification, the process by which intelligence develops (Prokopenko et al. 2025). The main author of this agenda (Mariana Meireles) is currently working on the problem of defining the relationship between entropy and the development of capabilities within LLMs. Urbina-Rodriguez et al. (2026) show a relationship between model capability and synergy, the information that a set of units carries jointly but none of them carries alone (Williams & Beer 2010), the same kind of information processing that the brain relies on for

complex cognition (Luppi et al. 2022; Proca et al. 2024). In work-in-progress, unpublished research she extended their result and found that the correlation also seems to hold at a neuron level of granularity.

3.1 Limits of Information Theory

However, information theory on its own is obviously not enough when exploring intelligence. It lacks many nuances that are broadly contested, for example by the critics of the strong version of Integrated Information Theory (IIT) (Aaronson 2014; Doerig et al. 2019). Even weak versions of IIT (Mediano et al. 2022; Barrett et al. 2026) do not address all the critics satisfyingly (Leung & Tsuchiya 2023). The reason is that information theory is intrinsically missing the tools to explain such a broad concept as intelligence.

In biology one has many examples of how complexification doesn't explain the whole picture. Some extremely simple systems have very complex behavior. Rule 110, a one-dimensional cellular automaton defined by eight rules, can compute anything a Turing machine can (Cook 2004). Distilled models are another example of regression of complexity but maintenance of intelligence (Hinton, Vinyals & Dean 2015).

4 Causal Inference

We believe that including causal inference in the exploration of intelligence is fundamental to this framework. We don't only want to have an abstract measurement of emergence, we also want the phenomenal effects of these capabilities in the world. Hoel & Levin (2020) have done this for gene regulatory networks and play an important inspirational role, as well as Barnett with Granger causality but we also believe that the current state-of-the-art is not enough.

The current state of the art in causal inference assumes a fixed causal structure between variables (Pearl 2009). It is not well suited to systems that reach the same outcome through different paths, which is exactly our definition of intelligence. When many paths lead to the same goal, the paths then seem epiphenomenal, and intervening on the path a system happened to take tells us very little about the capability we observe. Biologists call this *degeneracy* (Edelman & Gally 2001). One example is so-called junk DNA. Ohno (1972) used the term for the large part of the genome that does not code for proteins and seemed to have no function. We now know that much of it is regulatory (ENCODE Project Consortium 2012), although how much is still debated (Graur et al. 2013). Some of these non-coding regions are necessary for genes to be expressed where and when they should. Deleting elements of the junk or epiphenomenal DNA destroys capabilities we are able to observe (Sagai et al. 2005). We are not sure how to tackle this issue yet, but two avenues seem reasonable to us. The first is causal abstraction and causal emergence (Beckers & Halpern 2019; Geiger et al. 2025; Hoel, Albantakis & Tononi 2013).

5 Goals for the Residency

During the residency we intend to explore these questions to the point of being able to develop a robust research agenda and start a research institute that makes real progress on them. We hope that, with the help of the melting pot of fields this agenda currently draws from and will draw from in the future, we can succeed in mapping the manifold of intelligent systems.

6 Relevance to AI Safety

We see understanding the principles of intelligence as one of the most valuable moonshot projects for AI safety. What we understand we can measure, and what we can measure we can intervene

on. With these principles in hand we can change how models are trained and steer the design of AI architectures towards more interpretable constructions, so that systems built this way are both more capable and more aligned with humanity's flourishing.

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